Sharp Finite Markov Order
in Intrinsic Sofic Dimension

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Abstract

Let n be the real Hankel dimension of the cylinder probabilities of a stationary finite-alphabet process. If its Markov order is finite, we prove that it is at most $\binom{n}{2}$. For every $n \geq 2$, we construct a stationary process with a nonnegative rational presentation of minimal dimension n and exact order $\binom{n}{2}$, allowing the alphabet to grow. In the stationary sofic setting this improves the horizon 2^{n^2-1} of Béal, Jugé, Mairesse and Perrin, *Sofic measures* (2026), Theorem 5.1. The deterministic binomial-delay precedent is due to Béal and Senellart (1998); the extremal pair-chain core occurs in Trahtman's 1998 construction, which we realize stochastically. The upper proof combines Holland's rank criterion with classical exterior-square and nilpotence arguments. An accompanying Lean development proves the one-sided process theorem and rational stationary sharpness.

2 The result and its scope

Finite memory is a property of the output law, whereas a hidden-state presentation can contain redundant states. A representation-independent bound should therefore use the dimension of the word Hankel span. Our upper theorem applies to every stationary law with finite real Hankel dimension. The sharp examples have nonnegative rational presentations and are sofic in the usual finite-state sense.

Let A be a finite alphabet and μ a shift-invariant probability measure on $A^{\mathbb{N}}$, with its discrete product measurable structure. For a finite word w, put

$$p(w) = \mu\{x : x_0 x_1 \cdots x_{|w|-1} = w\}, \qquad p(\epsilon) = 1.$$

Write uv for concatenation and ϵ for the empty word. Cylinder probabilities satisfy

$$p(w) \geq 0, \qquad \sum_{a \in A} p(wa) = p(w) = \sum_{a \in A} p(aw). \tag{1}$$

The last equality uses stationarity.

Definition 2.1. The law is k-step Markov, for $k \geq 0$, if, for all words u, v, z with $|v| = k$ and $p(uv) > 0$,

$$\frac{p(uvz)}{p(uv)} = \frac{p(vz)}{p(v)}. \tag{2}$$

Its finite Markov order is the least such k, if one exists. Order zero means that the law is IID.

Left consistency implies $p(v) > 0$ whenever $p(uv) > 0$. This definition compares all finite future blocks; it makes no reference to matrix ranks.

For $w \in A^*$, let $c_w(u) = p(uw)$, and define

$$\mathcal{H}_p = \operatorname{span}_{\mathbb{R}}\{c_w : w \in A^*\}, \qquad n = \dim_{\mathbb{R}} \mathcal{H}_p.$$

Theorem 2.2. *If μ is stationary, $n < \infty$, and μ has finite Markov order, then*

$$\operatorname{ord}(\mu) \leq \binom{n}{2}.$$

For every $n \geq 2$, there is a stationary sofic law with a nonnegative rational presentation, real Hankel dimension n, and exact Markov order $\binom{n}{2}$. One construction uses $\binom{n}{2} - 1 + n^2$ output letters.

No fixed-alphabet optimum is asserted. A probability law has $n \geq 1$; when $n = 1$, the formula gives order zero.

2.1 Prior work and conventions

Béal–Jugé–Mairesse–Perrin [1, Theorem 5.1] give the horizon 2^{n^2-1} in the same minimal real representation dimension for invariant sofic measures. Their Theorem 4.4 attributes the rank criterion to Holland [3]. The original Holland scan was unavailable in the recorded check; the later reproduction was inspected. The needed word-law criterion is proved below.

Béal–Senellart [2, Proposition 3] already obtained the deterministic local-automaton delay $N(N-1)/2$, with N counting presentation states. Their Proposition 9 gives a binary quadratic-delay family. Trahtman's construction [6, Section 6] contains the lexicographic rank-two pair chain used in our sharpness proof. Neither the deterministic binomial phenomenon nor this combinatorial core is claimed as an invention of the project. The contribution considered here is the joined intrinsic stochastic theorem, rational stationary realization, minimality and exact order.

The source [1] uses two-sided invariant measures. Restriction to nonnegative coordinates preserves all word probabilities, Hankel dimension and (2). Conversely, (1) gives consistent probabilities on finite intervals of $\mathbb{Z}$; the finite-alphabet extension theorem gives a stationary two-sided law with those word probabilities. The conditional-block convention is preserved. This standard extension explains the comparison; the Lean endpoints themselves are on $\mathbb{N}$, and no separately formalized two-sided extension is claimed.

3 Minimal representations and the rank criterion

A finite real representation consists of a finite-dimensional space V, operators T_a, a vector g, and a functional l, such that

$$p(w) = lT_w g, \quad T_{a_1 \cdots a_m} = T_{a_1} \cdots T_{a_m}, \quad T_\epsilon = I.$$

It is reduced if the $T_w g$ span V and the lT_u separate its points. In finite dimension the latter condition says that the rows span V^*.

Lemma 3.1. *The finite-dimensional space $\mathcal{H}_p$ supplies a reduced representation. Every finite representation has dimension at least n, and every reduced one has dimension n.*

Proof. On functions of words let $(T_a f)(u) = f(ua)$. Since $T_a c_w = c_{aw}$, these maps preserve $\mathcal{H}_p$. Take $g = c_\epsilon$ and $l(f) = f(\epsilon)$. Then $T_w g = c_w$ and $lT_w g = p(w)$. Reachability follows, and $lT_u f = f(u)$ proves observability.

For any representation, the map $x \mapsto (u \mapsto lT_u x)$ sends $T_w g$ to c_w. Its restriction to the reachable space is onto $\mathcal{H}_p$, giving the dimension lower bound. In a reduced representation it is also injective, hence an isomorphism. □

Proposition 3.2 (Holland's rank criterion)**.** *For a reduced representation of a stationary word law, (2) holds at order k if and only if every length-k matrix T_v has rank at most one.*

Proof. Suppose (2) holds and $|v| = k$. If $p(v) = 0$, consistency and nonnegativity give $p(uvz) = 0$ for all u, z. Spanning rows and columns imply $T_v = 0$.

If $p(v) > 0$, the Markov identity gives

$$p(v)p(uvz) = p(uv)p(vz).$$

It remains true when $p(uv) = 0$, since right consistency forces $p(uvz) = 0$. Spanning the rows and columns gives

$$p(v)T_v = (T_v g)(lT_v),$$

so T_v has rank at most one.

Conversely, write a rank-one matrix as $T_v = xy$, with x a column and y a row. Commuting the resulting scalar factors gives

$$(lT_u T_v T_z g)(lT_v g) = (lT_u T_v g)(lT_v T_z g).$$

For a positive-mass history uv, also $p(v) > 0$, and division proves (2). A zero T_v admits no such history. □

This includes forbidden words and $k = 0$. No entrywise positivity of a reduced real representation is required.

4 The exterior-square cutoff

Lemma 4.1 (Uniform nilpotence)**.** *Let $\mathcal{T}$ be a family of endomorphisms of a d-dimensional vector space. If every product of some fixed length is zero, then every product of length d is zero.*

Proof. Let K_j be the common kernel of all length-j products, with $K_0 = 0$. Then

$$K_j \subseteq K_{j+1}, \qquad K_{j+1} = \bigcap_{T \in \mathcal{T}} T^{-1}(K_j).$$

A plateau is therefore permanent. Since some K_m is the full space, the chain cannot stabilize at a proper subspace. There are at most d strict dimension increases from zero, so $K_d = V$. □

This elementary instance of classical nilpotent linear-semigroup theory is included for its precise dimension mechanism; it is not claimed as a new nilpotence principle.

Upper bound in Theorem 2.2. Use Lemma 3.1. Put $W = \Lambda^2 V$, $D = \dim W = \binom{n}{2}$, and $S_a = \Lambda^2 T_a$. Classical exterior-power identities give

$$\Lambda^2(AB) = (\Lambda^2 A)(\Lambda^2 B), \qquad \Lambda^2 A = 0 \iff \operatorname{rank} A \leq 1.$$

Finite Markov order and Proposition 3.2 make every product of some fixed length in the S_a's zero. Lemma 4.1 then makes every length-D product zero. The exterior criterion and Proposition 3.2 imply D-step Markovity. □

The hypothesis is universal over words of a given length. It is distinct from the existential zero-word condition of matrix mortality.

5 Rational stationary sharpness

Fix $n \geq 2$, put $D = \binom{n}{2}$, and enumerate all two-subsets as $I_1, \ldots, I_D$. For consecutive pairs $I_s = \{i < j\}$, $I_{s+1} = \{k < l\}$, set

$$A_s = E_{i,k} + E_{j,l} \qquad (1 \leq s < D).$$

This maps the coordinate pair I_{s+1} to I_s and kills all other coordinate vectors.

The inherited construction. For lexicographic enumeration, these are exactly the numerical rank-two transition matrices in Trahtman's Section 6 [6], under its source-row convention. Column action reverses the description of direction. His Lemma 6.2 exhibits the path through all D unordered pairs. The local-testability order there is a different predicate from Markov order here. Our identification uses the explicit letters; it does not import that paper's entire local-testability theorem chain.

In the exterior basis indexed by the I_s's,

$$\Lambda^2 A_s = E_{s,s+1}, \qquad \Lambda^2(A_1 \cdots A_{D-1}) = E_{1,D} \neq 0.$$

Every product of D adjacent matrix units vanishes. Thus all length-D chain products have rank at most one, while the critical ordered product has rank two. When $n = 2$, there are no A_s's and the critical product is I_2, again of rank two.

5.1 Positive completion and minimality

Let

$$P = \frac{1}{n} J_n, \qquad \varepsilon = \frac{1}{2nD}, \qquad R = P - \varepsilon \sum_{s=1}^{D-1} A_s.$$

Every entry of the sum is at most $D - 1$, so

$$R_{ij} \geq \frac{1}{n} - \frac{D-1}{2nD} = \frac{D+1}{2nD} > 0.$$

Use letters a_s, $s < D$, and b_{ij}, $1 \leq i, j \leq n$, with

$$T_{a_s} = \varepsilon A_s, \qquad T_{b_{ij}} = R_{ij} E_{ij}.$$

The entries are nonnegative rational and $\sum_a T_a = P$. Take

$$l = \tfrac{1}{n}(1,\dots,1), \qquad g = \mathbf{1}.$$

Since $lP = l$, $Pg = g$, and $lg = 1$, the resulting $p(w) = lT_w g$ satisfies (1). The single-entry fillers give

$$T_{b_{ij}} g = R_{ij} e_i, \qquad lT_{b_{ij}} = \frac{R_{ij}}{n} e_j^{\mathsf{T}}.$$

Their positive coefficients prove reachability and observability, including $n = 2$. Thus the Hankel dimension is exactly n. These fillers agree with the formal construction.

5.2 An actual stationary process

For every state pair (i,j), set $\pi_{ij}(a) = n(T_a)_{ij}$; this is a probability distribution. Choose a random function $F : \{1,\dots,n\}^2 \to A$ with independent coordinates having those distributions. Let $Z_t = (Q_t, F_t)$ be IID, with Q_t uniform and independent of F_t. Define

$$X_t = F_t(Q_t, Q_{t+1}).$$

This two-block factor of a finite IID source is stationary. It is sofic because (Z_t, Z_{t+1}) is a finite-state Markov chain and X_t is a deterministic function of its state. For $w = a_1 \cdots a_m$, conditioning on $Q_0, \dots, Q_m$ yields

$$\Pr(X_0 \cdots X_{m-1} = w) = \sum_{q_0,\dots,q_m} \frac{1}{n^{m+1}} \prod_{t=1}^{m} n(T_{a_t})_{q_{t-1},q_t} = lT_w g.$$

This supplies an actual probability process with the stated cylinders.

5.3 Exact order

Any word containing a filler has rank at most one. Pure chain words have rank at most one at length D. The rank criterion therefore gives D-step Markovity.

The critical word $a_1 \cdots a_{D-1}$ has rank two and positive mass

$$\frac{2}{n} \varepsilon^{D-1}.$$

For $n = 2$, this is the empty word and its mass is one. If order $k < D$ held, the length-k prefix matrix of the critical word would have rank at most one; multiplying by its remainder would contradict its rank two. This proves sharpness in Theorem 2.2.

6 Formal verification and provenance

The accompanying `lean/` directory supplies a self-contained development for this report, its pinned manifest, entrypoint `Publication.lean` and `Audit.lean`. The environment is Lean 4.33.1 and Mathlib commit 0df444a360eaa60ab8c11dca51a86af692955474, credited in [4, 5]. The included `verification/` supplement supplies build logs, exact statement and axiom audits, source hashes, and verification provenance; its `README.txt` explains reproduction and `SHA256SUMS` checks the proof files. The report-to-formal correspondence is recorded in `CLAIM_MAP.txt`. This revision extracts the needed lemmas into subject-specific modules; the supplement records their correspondence to the earlier proof and the fresh build checks.

The principal declarations, in namespace `SoficMarkovOrder`, are:

- `stationary_process_markov_cutoff_of_finite_hankel`;
- `stationary_process_markov_cutoff_of_representation`;
- `exists_rational_sharp_sofic_process`;
- `processMarkov_iff_rankOne`.

The endpoint axioms are limited to `propext`, `Classical.choice`, and `Quot.sound`. The upper endpoint constructs a reduced representation internally. The Trahtman comparison and two-sided extension explanation are report-level arguments, not separately formalized Lean theorems.

Documented research process. The AI investigation reformulated rank collapse on the exterior square, then exposed the uniform-nilpotence argument through common-kernel stabilization. Exact small-dimensional computations preceded the all-dimensional formal proof. Formalization replaced row fillers by single-entry residual letters and used a two-block IID construction for actual process semantics. Later primary-source inspection identified Trahtman's matching pair-chain core and narrowed the originality account.

Formal checking and the focused literature search do not certify global priority. No fixed-alphabet classification or efficiency claim for large constructions is asserted. Corrections should identify the version and affected statement through the route in the accompanying `CORRECTIONS.txt`.

References

[1] Marie-Pierre Béal, Vincent Jugé, Jean Mairesse, and Dominique Perrin. Sofic measures. arXiv:2604.11212v1, 2026. 13 April 2026. Theorems 4.4 and 5.1. https://arxiv.org/abs/2604.11212v1.

[2] Marie-Pierre Béal and Jean Senellart. On the bound of the synchronization delay of a local automaton. *Theoretical Computer Science*, 205(1–2):297–306, 1998. Propositions 3 and 9. https://hal.science/hal-00619330.

[3] Paul W. Holland. Some properties of an algebraic representation of stochastic processes. *The Annals of Mathematical Statistics*, 39(1):164–170, 1968. Rank criterion inspected through the reproduction in Béal et al. (2026). https://doi.org/10.1214/aoms/1177698514.

[4] Lean project contributors. Lean theorem prover, version 4.33.1. Software, 2026. https://github.com/leanprover/lean4/releases/tag/v4.33.1.

[5] The mathlib community. The mathlib4 mathematical library. Software, pinned revision in the accompanying Lake manifest, 2026. https://github.com/leanprover-community/mathlib4.

[6] A. N. Trahtman. Precise estimation on the order of local testability of deterministic finite automaton. In *Automata Implementation*, volume 1436 of *Lecture Notes in Computer Science*, pages 198–212. Springer, 1998. Section 6, especially Lemma 6.2. https://u.cs.biu.ac.il/~trakhta/lecnotes.pdf.